NEFT Banking Transaction Analysis

Project Report — End-to-End Data Analytics Case Study

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1. Executive Summary

This project analyzes 12+ years of India's NEFT (National Electronic Funds Transfer) banking data, covering 267 banks and over 21,000 bank-month records between June 2008 and June 2020. The goal was to build a full analytics pipeline — from raw data cleaning through SQL-based analysis to an interactive Power BI dashboard — to understand how NEFT adoption grew over time, which banks dominate transaction volume, and what seasonal or structural patterns exist in the data.

Key headline numbers:

Metric	Value
Total Debit Amount	1,043,992,298 Cr
Total Credit Amount	1,044,026,051 Cr
Total Debit Transactions	12,900,081,237
Total Credit Transactions	12,899,866,676
Total Amount (Debit + Credit)	2,088,018,349 Cr
Total Transactions (Debit + Credit)	25,799,947,913
Unique Reporting Banks	267
Date Range	Jun 2008 - Jun 2020

2. Methodology

The project follows a standard analytics-engineering workflow, mirrored in the folder structure of this repository:

- **Data Cleaning (Python & SQL):** standardized bank names, parsed the 'Month YYYY' text field into proper dates, filled missing numeric values, removed duplicate bank/month records, and derived total transaction/amount fields.
- **Database Design (MySQL):** a staging table mirrors the raw CSV; a cleaned production table (neft_transactions) holds analysis-ready data; an optional star-schema (dim_bank / fact_neft_monthly) supports BI-tool style modelling.
- **Feature Engineering (Python):** month-over-month and year-over-year growth rates, 3-month rolling averages, market share per month, overall bank ranking, and debit/credit ratios.
- **Exploratory Data Analysis (Python):** summary statistics and trend/distribution charts (see images/ folder).
- **Dashboard (Power BI):** an interactive report with bank/year/month slicers, KPI cards, and trend/composition visuals (see powerbi/NEFT_Analysis.pbix and dashboard/dashboard.png).

3. Key Findings

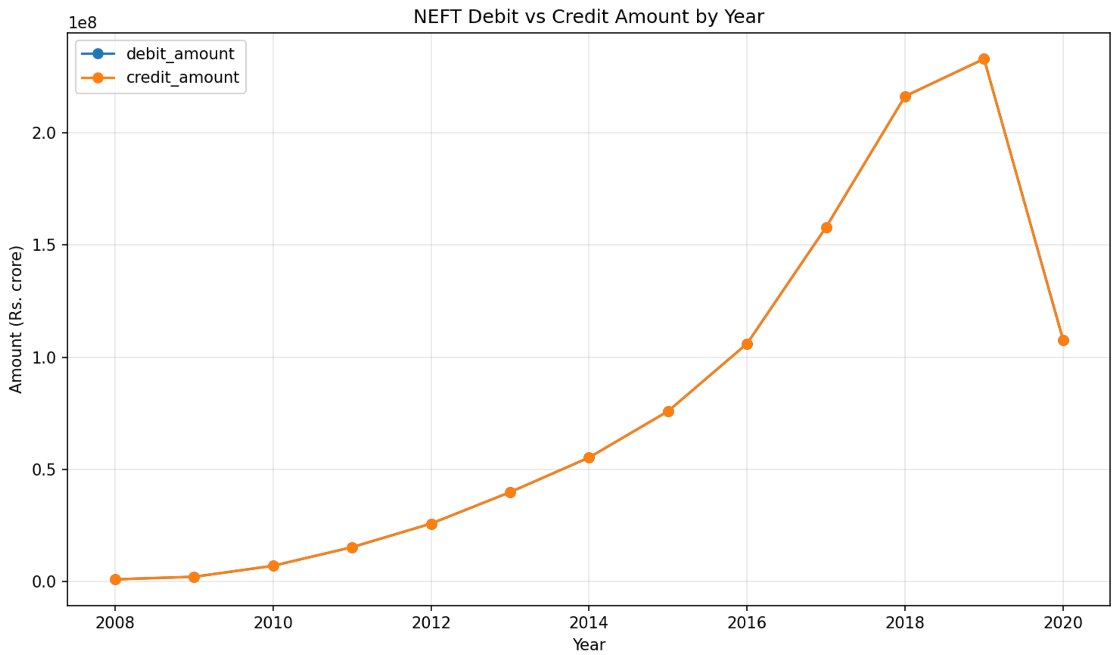
3.1 Market Concentration

NEFT transaction value is highly concentrated: the top 5 banks (State Bank of India, HDFC Bank, ICICI Bank, Axis Bank, and Citi Bank) account for approximately **48.6%** of total NEFT amount across the full 2008-2020 period, while the top 10 banks account for roughly 64%. This reflects the outsized role of India's largest public-sector and private-sector banks in digital payment infrastructure.

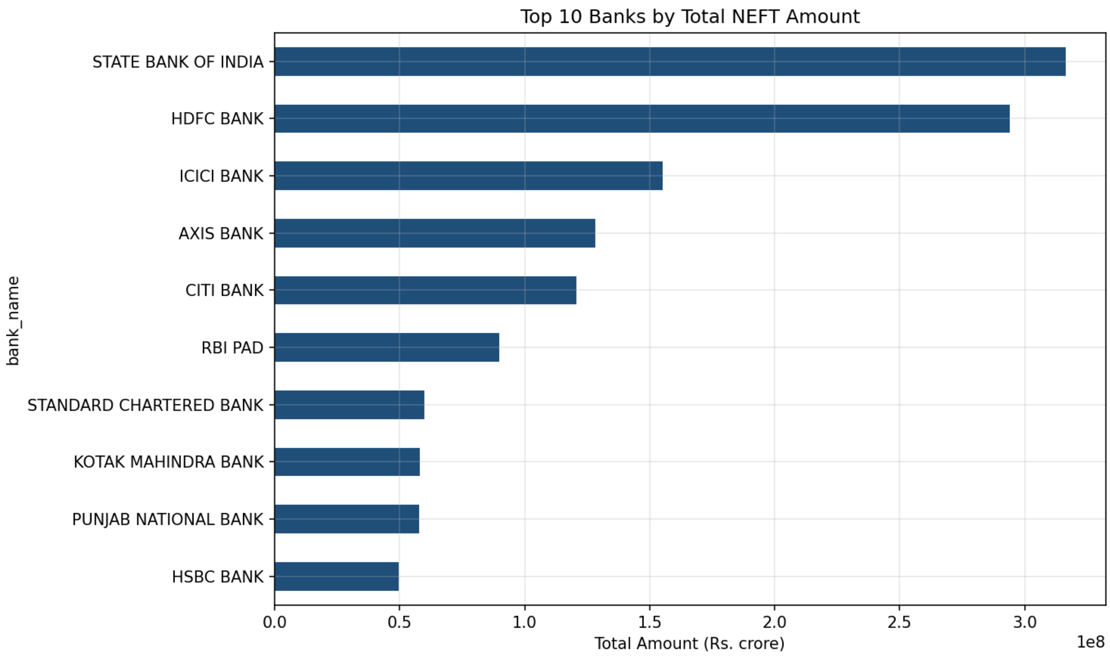
Rank	Bank	Total Amount (Cr)	Market Share
1	State Bank Of India	316,518,985	15.16%
2	Hdfc Bank	293,955,817	14.08%
3	Icici Bank	155,081,441	7.43%
4	Axis Bank	128,314,188	6.15%
5	Citi Bank	120,602,836	5.78%
6	Rbi Pad	89,943,723	4.31%
7	Standard Chartered Bank	59,977,120	2.87%
8	Kotak Mahindra Bank	58,118,257	2.78%
9	Punjab National Bank	57,641,220	2.76%
10	Hsbc Bank	49,760,357	2.38%

3.2 Growth Trend

Total NEFT transaction amount grew from roughly 2.1 million Cr in 2008 to over 465 million Cr in 2019, before a partial dip in 2020 (data through June only, coinciding with COVID-19 disruption). Between 2009 and 2019, the compound annual growth rate (CAGR) of total NEFT amount was approximately **59.5% per year**, reflecting rapid digitization of retail and corporate payments in India over the decade.

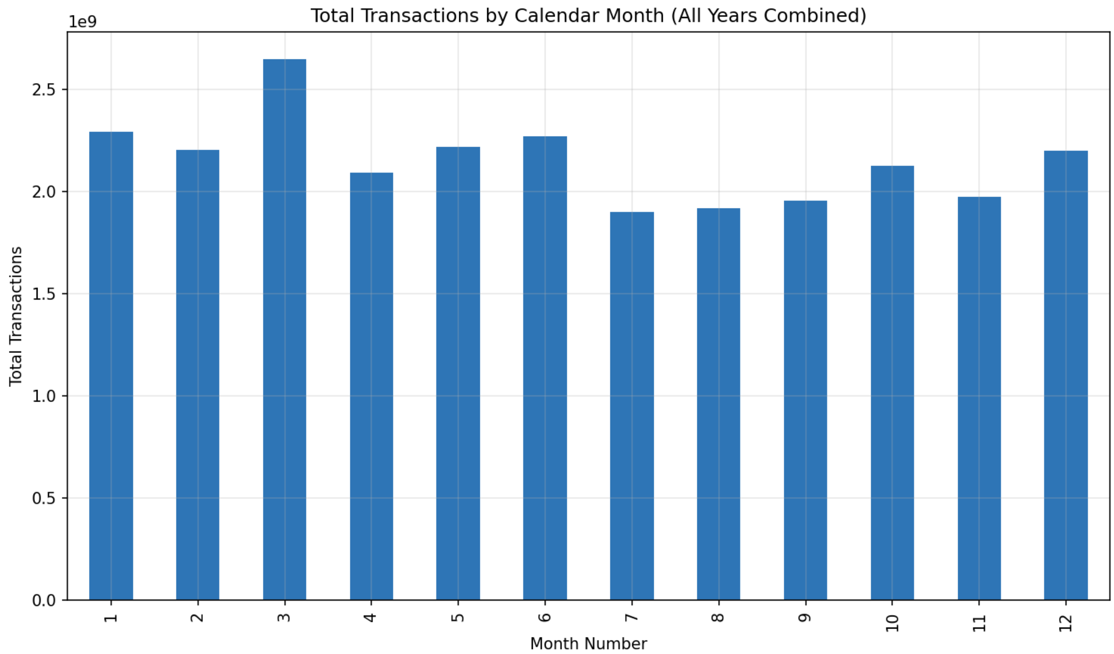


3.3 Top Banks by Volume



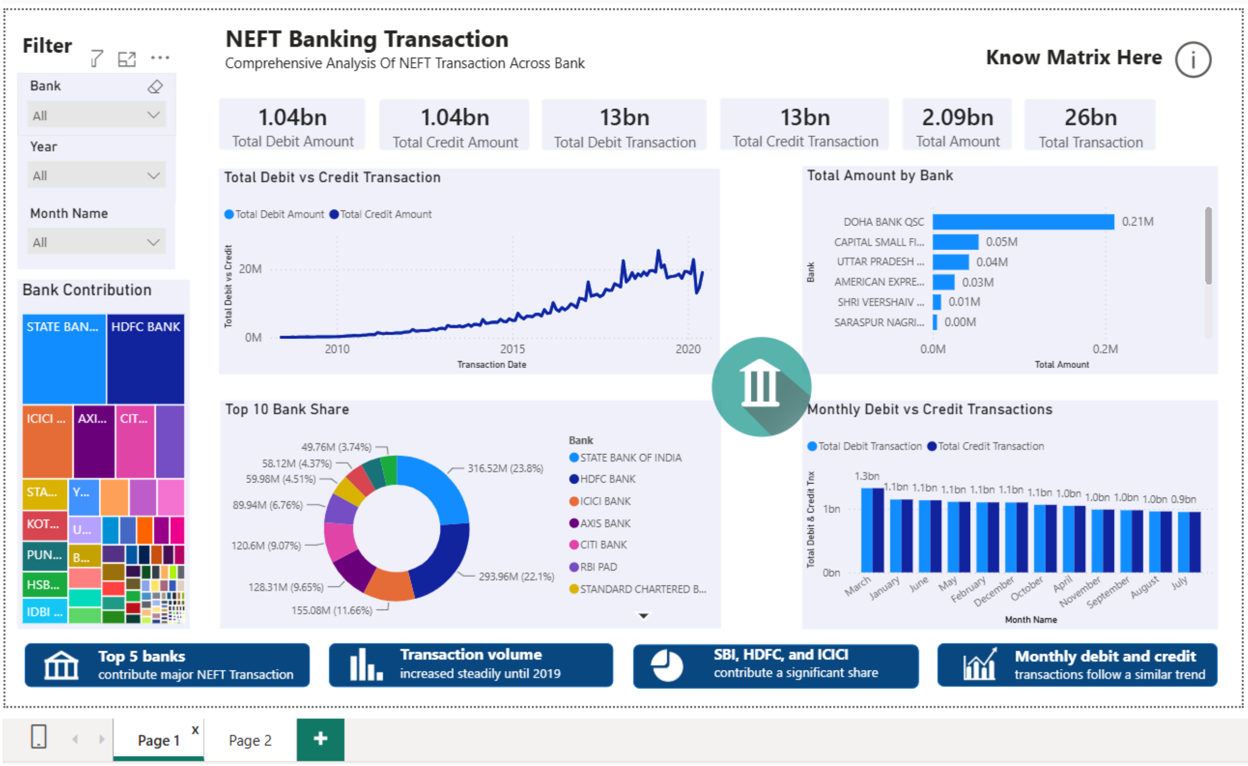
3.4 Seasonality

Aggregating transaction counts by calendar month (across all years) shows month-to-month variation consistent with typical business and fiscal-year-end payment cycles, with March (the end of the Indian financial year) among the busier months. This is a useful signal for banks and payment infrastructure teams when planning capacity.



4. Power BI Dashboard

The interactive dashboard (powerbi/NEFT_Analysis.pbix) exposes the same data through KPI cards, a debit-vs-credit trend line, a bank-contribution treemap, a top-10 donut chart, and a monthly debit-vs-credit column chart, all filterable by bank, year, and month.



5. Tech Stack

- **Python** (pandas, numpy, matplotlib) — data cleaning, feature engineering, EDA
- **MySQL** — schema design, cleaning transforms, analytical SQL (window functions, CTEs)
- **Power BI** — interactive dashboard and DAX-based KPI cards
- **Git / GitHub** — version control and project hosting

6. Conclusion & Next Steps

This analysis confirms strong, sustained growth in NEFT adoption over 2008-2019 and a market structure dominated by a handful of large banks. Potential extensions include: forecasting future transaction volume with a time-series model, joining in RTGS/IMPS/UPI data for a full digital-payments comparison, and building bank-level anomaly detection to flag unusual month-over-month swings.